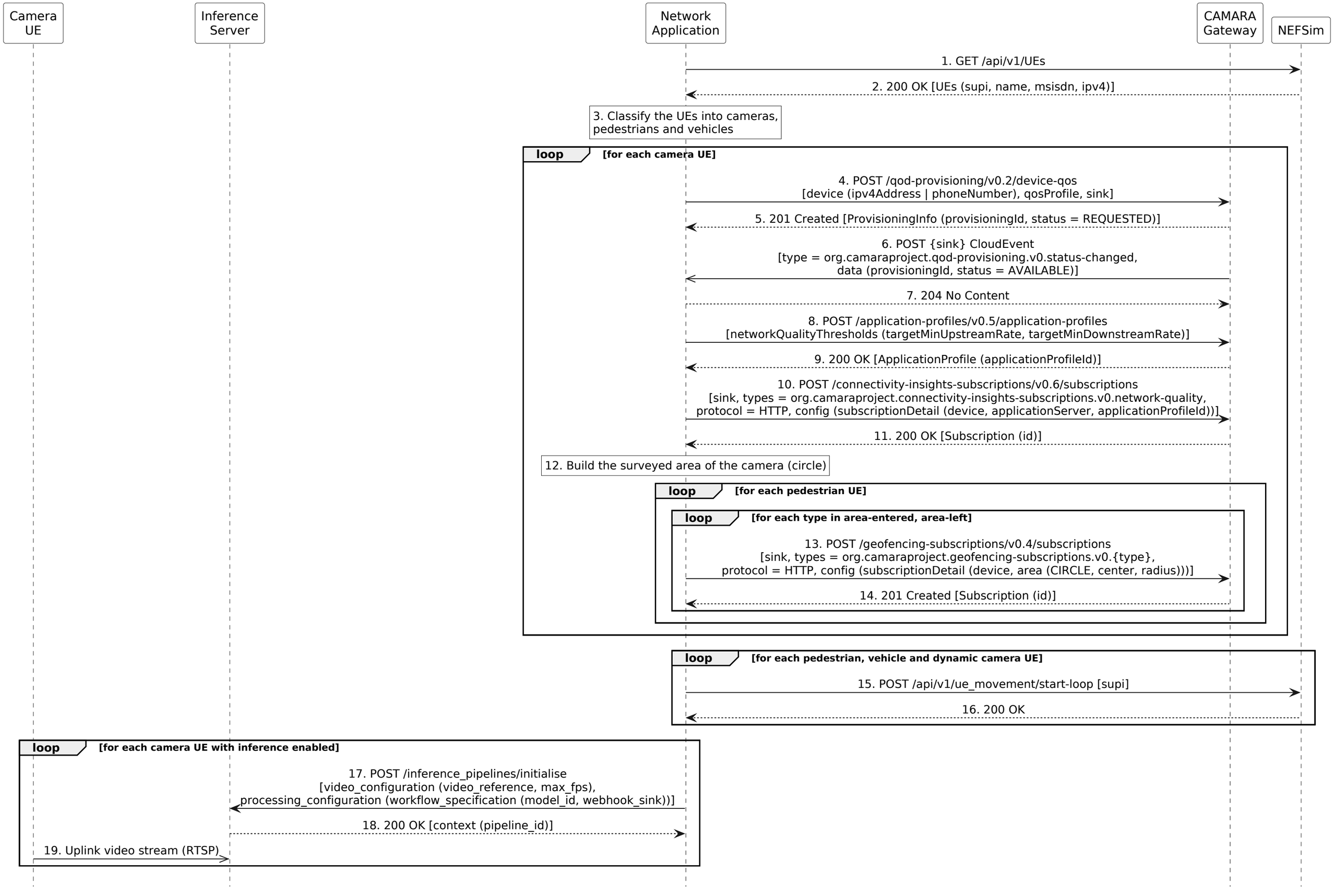
Camera UE
Inference Server
Network Application
CAMARA Gateway
NEFSim
1. GET /api/v1/UEs
2. 200 OK [UEs (supi, name, msisdn, ipv4)]
3. Classify the UEs into cameras, pedestrians and vehicles
loop [for each camera UE]
4. POST /qod-provisioning/v0.2/device-qos [device (ipv4Address | phoneNumber), qosProfile, sink]
5. 201 Created [ProvisioningInfo (provisioningId, status = REQUESTED)]
6. POST {sink} CloudEvent [type = org.camaraproject.qod-provisioning.v0.status-changed, data (provisioningId, status = AVAILABLE)]
7. 204 No Content
8. POST /application-profiles/v0.5/application-profiles [networkQualityThresholds (targetMinUpstreamRate, targetMinDownstreamRate)]
9. 200 OK [ApplicationProfile (applicationProfileId)]
10. POST /connectivity-insights-subscriptions/v0.6/subscriptions [sink, types = org.camaraproject.connectivity-insights-subscriptions.v0.network-quality, protocol = HTTP, config (subscriptionDetail (device, applicationServer, applicationProfileId))]
11. 200 OK [Subscription (id)]
12. Build the surveyed area of the camera (circle)
loop [for each pedestrian UE]
loop [for each type in area-entered, area-left]
13. POST /geofencing-subscriptions/v0.4/subscriptions [sink, types = org.camaraproject.geofencing-subscriptions.v0.{type}, protocol = HTTP, config (subscriptionDetail (device, area (CIRCLE, center, radius)))]
14. 201 Created [Subscription (id)]
loop [for each pedestrian, vehicle and dynamic camera UE]
15. POST /api/v1/ue_movement/start-loop [supi]
16. 200 OK
loop [for each camera UE with inference enabled]
17. POST /inference_pipelines/initialise [video_configuration (video_reference, max_fps), processing_configuration (workflow_specification (model_id, webhook_sink))]
18. 200 OK [context (pipeline_id)]
19. Uplink video stream (RTSP)